

## Session :- 3

**1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.**

**Ans:**

| Abbreviation | Full Form                          | One-line Use                                                                            |
|--------------|------------------------------------|-----------------------------------------------------------------------------------------|
| <b>HTTP</b>  | HyperText Transfer Protocol        | Used to transfer web pages between a web browser and a web server.                      |
| <b>HTTPS</b> | HyperText Transfer Protocol Secure | Used to securely transfer data between a web browser and a web server using encryption. |
| <b>FTP</b>   | File Transfer Protocol             | Used to upload and download files between computers over a network.                     |
| <b>SMTP</b>  | Simple Mail Transfer Protocol      | Used to send emails from one mail server to another.                                    |

**2. What is the difference between IPv4 and IPv6? Answer in 2 lines.**

**Ans:** IPv4 uses 32-bit addresses (e.g., 192.168.1.1) and supports fewer unique IP addresses.

IPv6 uses 128-bit addresses (e.g., 2001:db8::1) and provides a much larger number of unique IP addresses.

### 3. Run nslookup google.com on your system and write down the IP address you get.

**Ans:** I can't run nslookup on your own system or see its output.

To get the IP address, run this command in your terminal or Command Prompt:

```
nslookup google.com
```

Example output (this may be different on your system):

Name: google.com

Address: 142.250.190.14

IP address (example): 142.250.190.14

**Note:** The IP address returned by nslookup google.com can vary depending on your location, DNS server, and Google's load balancing.

### 4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

**Ans:**

| HTTP Status Code      | Meaning                                                                     |
|-----------------------|-----------------------------------------------------------------------------|
| 200 OK                | The request was successful, and the server returned the requested resource. |
| 301 Moved Permanently | The requested resource has been permanently moved to a new URL.             |
| 403 Forbidden         | The server understands the request but refuses to                           |

| HTTP Status Code          | Meaning                                                                  |
|---------------------------|--------------------------------------------------------------------------|
|                           | allow access.                                                            |
| 404 Not Found             | The requested page or resource could not be found on the server.         |
| 500 Internal Server Error | The server encountered an unexpected error while processing the request. |

## 5. Explain DNS hierarchy (root, TLD, domain, subdomain) with **www.google.com** as an example.

**Ans:** The DNS (Domain Name System) hierarchy is organized from the most general level to the most specific:

- Root (.) – The top level of the DNS hierarchy, represented by a dot (.), which directs queries to the correct Top-Level Domain (TLD).
- TLD (Top-Level Domain) – The last part of the domain name, such as .com.
- Domain – The registered name under the TLD, such as google in google.com.
- Subdomain – A subdivision of the main domain, such as www in www.google.com.

Example: www.google.com

- Root: .
- TLD: com
- Domain: google
- Subdomain: www

